

ML Data Dictionary

Aegis Entropy: AI-Based Intelligent System for Network Security Monitoring & Automated Response

Module	Artificial Intelligence
Attack Focus	Port Scanning & Network Reconnaissance
Status	Academic Project (Severity: MEDIUM-HIGH)
Version	Aegis Entropy (Final Update)

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Network Features Data Dictionary

This table details the network features extracted and used by the Machine Learning model (Random Forest / XGBoost / Isolation Forest) for detection and active response. It integrates the structural updates of Aegis Entropy (Shadow Nodes and TCP Tarpitting).

Feature Name	Type	Description & Relevance
Distinct Dst Ports/IP	Numeric	Number of unique destination ports contacted by the source. <i>Relevance: Critical-core scanning signature.</i>
SYN Flag Count	Numeric	SYN packets sent without completing the handshake. <i>Relevance: Critical stealth SYN scan indicator.</i>

Feature Name	Type	Description & Relevance
RST Flag Count	Numeric	RST responses from closed ports. <i>Relevance: Very High-port probe rejection indicator.</i>
Flow Duration	Numeric	Duration of each individual probe connection. <i>Relevance: High-usually very short in fast scans.</i>
Total Fwd Packets	Numeric	Packets per flow from the scanner. <i>Relevance: Usually 1 packet per probe.</i>
ACK Flag Count	Numeric	ACK packets sent without a prior SYN. <i>Relevance: Medium-ACK scan detection.</i>
IAT Mean	Numeric	Mean inter-arrival time between probes. <i>Relevance: High-slow scans have a large IAT.</i>
Unique Dst IPs/Src	Numeric	Number of distinct destination hosts contacted. <i>Relevance: Medium-indicates host discovery phase.</i>
TTL Value	Numeric	Time-to-Live field in the IP header. <i>Relevance: Low-OS fingerprinting attempts.</i>
Bwd Packet Length	Numeric	Response packet size. <i>Relevance: Medium-differentiates RST vs SYN-ACK ratio.</i>
Shadow Node Interaction Updated	Binary (Numeric)	Indicates if the source IP contacted a decoy node (Inline Shadow Node). Replaces the old Honeypot Interaction Flag. <i>Relevance: CRITICAL-high-confidence threat signal.</i>
MTD Port Delta	Numeric	Difference between the probed port and the currently assigned port by mutation. <i>Relevance: High-reveals the use of stale reconnaissance data.</i>

Feature Name	Type	Description & Relevance
TCP Window Size (outbound) New	Numeric	Window size sent in the attacker-received SYN-ACK. <i>Relevance: New metric to evaluate Tarpit effectiveness.</i>